

HDOC DAY – 22

Algorithm:

Step 1: Start.

Step 2: Declare integer variables m, n, choice, i, and j.

Step 3: Input two positive integers m and n.

Step 4: Check whether the inputs are valid:

- If $m \leq 0$ or $n \leq 0$, display "Error: m and n must be positive integers."
- If $m > n$, display "Error: m must be less than or equal to n."
- Otherwise, continue.

Step 5: Display the menu

Step 6: Input the user's choice.

Step 7: Use a switch statement to perform the selected operation.

- Case 1: Use a loop from 1 to 10 and print the multiplication tables of m and n side by side.
- Case 2: Print column headings from 1 to n. Use nested loops to calculate and print the multiplication grid from 1 to m and 1 to n.
- Case 3: Use an outer loop from m to n. For each integer, use an inner loop to print that integer and the next five consecutive integers.
- Case 4: Display "Exiting program..." and terminate the menu loop.
- Default: Display "Invalid choice. Please enter a choice from 1 to 4."

Step 8: Repeat Steps 5–7 until the user selects option 4.

Step 9: Stop.

C PROGRAM:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int m, n, choice;
```

```
int i, j;
```

```
/* Input m and n */
```

```
printf("Enter the value of m: ");
```

```
scanf("%d", &m);
```

```
printf("Enter the value of n: ");
```

```
scanf("%d", &n);
```

```
/* Validate m and n */
```

```
if (m <= 0 || n <= 0)
```

```
{
```

```
    printf("Error: m and n must be positive integers.\n");
```

```
    return 0;
```

```
}
```

```
if (m > n)
```

```
{
```

```
    printf("Error: m must be less than or equal to n.\n");
```

```
    return 0;
```

```
}
```

```
/* Repeat menu until user selects Exit */
```

```
do
```

```
{
```

```
    printf("\n1. Multiplication tables of m and n\n");
```

```
    printf("2. Multiplication grid\n");
```

```
printf("3. Consecutive integers\n");
```

```
printf("4. Exit\n");
```

```
printf("Enter your choice: ");
```

```
scanf("%d", &choice);
```

```
switch (choice)
```

```
{
```

```
    case 1:
```

```
        /* Multiplication tables */
```

```
        printf("\nMultiplication Tables of %d and %d (m=%d, n=%d)\n\n",
```

```
            m, n, m, n);
```

```
        for (i = 1; i <= 10; i++)
```

```
        {
```

```
            printf("%d x %d = %-5d\t%d x %d = %d\n",
```

```
                m, i, m * i,
```

```
                n, i, n * i);
```

```
        }
```

```
        break;
```

```
    case 2:
```

```
        /* Multiplication grid */
```

```
        printf("\nMultiplication Grid (m=%d, n=%d)\n\n", m, n);
```

```
        /* Print column headings */
```

```
        printf("  ");
```

```
        for (j = 1; j <= n; j++)
```

```
{  
    printf("%4d", j);  
}
```

```
printf("\n  ");  
for (j = 1; j <= n; j++)  
{  
    printf("----");  
}  
printf("\n");
```

```
/* Print multiplication grid */
```

```
for (i = 1; i <= m; i++)  
{  
    printf("%2d | ", i);  
  
    for (j = 1; j <= n; j++)  
    {  
        printf("%4d", i * j);  
    }  
  
    printf("\n");  
}
```

```
break;
```

case 3:

```
/* Consecutive integers */  
printf("\nConsecutive Integers from %d to %d (m=%d, n=%d)\n\n",
```

```
    m, n, m, n);
```

```
    for (i = m; i <= n; i++)
```

```
    {
```

```
        printf("%d: ", i);
```

```
        for (j = i; j <= i + 5; j++)
```

```
        {
```

```
            printf("%d ", j);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    break;
```

```
case 4:
```

```
    printf("\nExiting program...\n");
```

```
    break;
```

```
default:
```

```
    printf("\nInvalid choice. Please enter a choice from 1 to 4.\n");
```

```
}
```

```
} while (choice != 4);
```

```
return 0;
```

```
}
```

```

Session Actions Edit View Help
#include <stdio.h>

int main()
{
    int m, n, choice;
    int i, j;

    /* Input m and n */
    printf("Enter the value of m: ");
    scanf("%d", &m);

    printf("Enter the value of n: ");
    scanf("%d", &n);

    /* Validate m and n */
    if (m <= 0 || n <= 0)
    {
        printf("Error: m and n must be positive integers.\n");
        return 0;
    }

    if (m > n)
    {
        printf("Error: m must be less than or equal to n.\n");
        return 0;
    }

    /* Repeat menu until user selects Exit */
    do
    {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                /* Multiplication tables */
                printf("\nMultiplication Tables of %d and %d (m=%d, n=%d)\n\n",
                    m, n, m, n);

                for (i = 1; i <= 10; i++)
                {
                    printf("%d x %d = %-5d\t%d x %d = %d\n",
                        m, i, m * i,
                        n, i, n * i);
                }
                break;

            case 2:
                /* Multiplication grid */
                printf("\nMultiplication Grid (m=%d, n=%d)\n\n", m, n);

                /* Print column headings */
                printf("    ");
                for (j = 1; j <= n; j++)
                {
                    printf("%4d", j);

```

```

/* Multiplication grid */
printf("\nMultiplication Grid (m=%d, n=%d)\n\n", m, n);

/* Print column headings */
printf(" ");
for (j = 1; j ≤ n; j++)
{
    printf("%4d", j);
}

printf("\n ");
for (j = 1; j ≤ n; j++)
{
    printf("——");
}
printf("\n");

/* Print multiplication grid */
for (i = 1; i ≤ m; i++)
{
    printf("%2d | ", i);

    for (j = 1; j ≤ n; j++)
    {
        printf("%4d", i * j);
    }

    printf("\n");
}
break;

case 3:
/* Consecutive integers */
printf("\nConsecutive Integers from %d to %d (m=%d, n=%d)\n\n",
    m, n, m, n);

for (i = m; i ≤ n; i++)
{
    printf("%d: ", i);

    for (j = i; j ≤ i + 5; j++)
    {
        printf("%d ", j);
    }

    printf("\n");
}
break;

case 4:
printf("\nExiting program... \n");
break;

default:
printf("\nInvalid choice. Please enter a choice from 1 to 4.\n");
}

} while (choice ≠ 4);

return 0;
}

```

Test case table:

Test case	m	n	choice	type	Expected result
1	3	5	1	valid	Tables of 3 and 5 from x1 to x10
2	3	5	2	Valid	Multiplication grid of 3 rows and 5 columns

3	3	5	3	Valid	Prints 3 to 8, 4 to 9, and 5 to 10
4	3	5	4	Valid	Program exits
5	3	5	7	Invalid choice	Displays invalid choice error
6	1	1	1	Boundary	Prints table of 1 twice, side by side
7	5	3	--	Invalid input	Error because $m > n$
8	0	5	--	Invalid input	Error because m is not positive

```

(kali㉿kali)-[~]
$ ./h22
Enter the value of m: 3
Enter the value of n: 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 3 and 5 (m=3, n=5)

3 x 1 = 3      5 x 1 = 5
3 x 2 = 6      5 x 2 = 10
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2

Multiplication Grid (m=3, n=5)

      1   2   3   4   5
    -----
1 |  1   2   3   4   5
2 |  2   4   6   8  10
3 |  3   6   9  12  15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3

Consecutive Integers from 3 to 5 (m=3, n=5)

3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

```